Digital Cinema Package (DCP) Quality Control (QC) Criteria

This document outlines the standard Pass and Fail conditions for the quality control testing of digital cinema packages. Each parameter must be evaluated prior to the final screening to ensure an optimal viewing and auditory experience. We test each category in turn and a failure in a category precipitates a failure overall.

The QC tests performed by Rich Mix are final checks to ensure your content will screen in the best manner possible in a cinema environment, we do not provide feedback on the editorial or artistic decision making. Each test lasts between half an hour to an hour, and a standardized feedback form will be sent once the tests are completed. Contact details must be provided for the QC checklist.

1. Ingestion & Integrity

Requirement: *The server must successfully ingest the entire package and validate the file hashes to ensure no data corruption occurred during the transfer process.*

- **PASS:**
    - The DCP ingests up to 100% without any server errors or interruptions.
    - The server's hash validation process completes successfully, confirming that all file sizes and checksums perfectly match the original package.
- **PASS WITH CAVEATS:**
    - The drive format requires manual intervention (e.g., an unsupported format requiring transfer via an intermediate staging drive), but the hashes ultimately validate.
    - Non-essential files or secondary CPLs (like an unused trailer) fail their hash checks, but the primary screening CPL validates and ingests perfectly.
- **FAIL:**
    - The ingestion process stalls, crashes, or produces an error message.
    - Hash validation fails, indicating that files are missing, incomplete, or corrupted during transfer.

2. KDM Validation

Requirement: *For encrypted content, the Key Delivery Message (KDM) must match the specific server's serial number and effectively unlock the Composition Playlist (CPL) for the designated screening time window. KDM's should be emailed to production@richmix.org.uk and must be unlocked for the agreed testing window.*

- **PASS:**

    - The KDM is successfully received at *production@richmix.org.uk*.
    - The KDM ingests without error, targets the correct projector/server serial number, and unlocks the content.
    - The valid time window covers both the agreed-upon testing period and the actual screening time.
- **PASS WITH CAVEATS:**
    - KDMs are unlocked for testing but not for the day of the screening.
    - The KDM expires unusually close to the end of the film's runtime, leaving no buffer for event delays.
- **FAIL:**
    - The KDM is not received, or is sent to the wrong email address.
    - The KDM fails to ingest due to a server serial number mismatch.
    - The content remains locked because the KDM has expired, is not yet active (outside the testing window), or is corrupted.

3. CPL Selection

Requirement: *If the package contains multiple CPLs, the correct version—such as 2D versus 3D, or specific language and subtitle configurations—must be identified and loaded.*

- **PASS:**
    - The correct CPL is clearly identifiable and successfully loaded into the show playlist.
    - The loaded CPL matches the exact specifications required for the screening (e.g., correct 2D/3D format, proper audio language, and intended subtitle track).
- **PASS WITH CAVEATS:**
    - Multiple ambiguously named CPLs are provided (e.g., `Feature_v1`, `Feature_Final_Edit`). The assumed correct version plays properly, but explicit client confirmation is needed to ensure the right cut is screened.
    - The CPL naming convention does not match the actual content (e.g., labeled as Scope but is actually Flat).
- **FAIL:**
    - An incorrect CPL is loaded (e.g., playing a 3D version on a 2D setup, or the wrong language version).
    - The CPL naming conventions are improperly formatted or too ambiguous to confidently determine which version is the correct one.

4. Visual Framing & Ratios

Requirement: *The aspect ratio must match the projector's macro settings, typically Flat (1.85:1) or Scope (2.39:1), ensuring the picture is framed correctly without cropping or misalignment with the screen masking.*

- **PASS:**
    - The content's aspect ratio perfectly aligns with the selected projector macro (Flat or Scope).
    - The image correctly fills the screen and aligns seamlessly with the physical screen masking, with no unintended cropping, stretching, or squeezing.
- **PASS WITH CAVEATS:**
    - The content uses a non-standard aspect ratio (e.g., 4:3 or 2.00:1) packaged within a standard Flat container. This leaves visible black bars (pillarboxing or letterboxing) on the screen, but the framing is intentional and no active picture is lost.
    - Minor masking bleed occurs due to slight discrepancies in the render, but it does not obscure essential action.
- **FAIL:**
    - The image is distorted (stretched or squeezed) due to a macro mismatch.
    - Essential visual information is cropped off the top, bottom, or sides by the physical screen masking.
    - Unintended heavy windowboxing or letterboxing occurs when the image should fill the screen.

5. Audio Mapping & Sync

Requirement: *Sound output must be tested to confirm the correct format (5.1 or 7.1) is routing to the appropriate channels, with precise lip-sync and standard overall volume levels.*

- **PASS:**
    - Audio is correctly routed according to its format (e.g., dialogue is isolated in the center channel; ambient effects are in the surrounds).
    - Picture and sound are in perfect synchronization (precise lip-sync).
    - The overall volume at the standard reference fader level is clear, comfortable, and free of distortion.
- **PASS WITH CAVEATS:**
    - The mix is exclusively Stereo (L/R) rather than 5.1/7.1, leaving the center and surround channels empty.
    - The overall mix is noticeably quiet or unusually loud, requiring the cinema fader to be adjusted significantly outside the standard range to achieve an appropriate listening level.
- **FAIL:**
    - Audio channels are swapped, missing, or improperly downmixed (e.g., dialogue coming from surround speakers or missing entirely).
    - There is a noticeable delay between the picture and the sound.
    - The audio is overly quiet, painfully loud, clipping, or features static/dropouts.

6. Subtitles & Accessibility

Requirement: *On-screen subtitles must render within the safe area without being cut off by masking, and closed captioning (CC) or audio description (AD) tracks must properly transmit to audience accessibility devices.*

- **PASS:**
    - On-screen subtitles are fully visible within the title safe area and are completely clear of the physical screen masking.
    - Accessibility devices (receivers/headsets) successfully pick up the CC and AD signals, and the tracks play in sync with the film.
- **PASS WITH CAVEATS:**
    - Subtitles are "burnt-in" to the video file rather than generated dynamically via XML, limiting adjustment options.
    - Subtitles sit unusually low and close to the masking edge, but remain completely legible from the auditorium.
- **FAIL:**
    - Subtitles bleed out of the frame or are cut off by the bottom/side masking, making them unreadable.
    - The CC/AD infrared or Wi-Fi emitters fail to transmit, or the tracks are missing, out of sync, or garbled on the testing devices.

7. Playback Stability

Requirement: *A visual check of high-motion or high-bitrate scenes ensures the server and projector handle the data throughput without dropped frames, stuttering, or visual artifacts.*

- **PASS:**
    - Playback remains smooth and consistent throughout the entire runtime.
    - High-action, high-detail, or high-bitrate scenes play flawlessly without any visual degradation.
- **PASS WITH CAVEATS:**
    - A minor, isolated visual artifact (e.g., slight color banding in dark scenes or a single dropped frame) is observed, but it appears to be "baked into" the source master file rather than a server playback error.
- **FAIL:**
    - The picture exhibits noticeable stuttering, jitter, or dropped frames.
    - Visual artifacts such as macroblocking (pixelation), color flashing, or screen tearing (when pieces of two or more different video frames are played in a single screen refresh).
    - The playback freezes or the server crashes during playback.